

Day 15 - Gateway Native Implementation Decision 100% Closure

Date: 2026-05-09

Branch: to-knowledge-mc

Lane: Gateway Native Implementation Decision

Status: DEVELOPER-SIDE CLOSED

Blocker class: OWNER_GATED for authenticated owner visual confirmation

Code commit: 25f577e

Rollback: git revert 25f577e

Scope

Day 15 decides and proves the Gateway implementation path.

Decision: keep the accepted FULL v3 designer HTML mounted directly inside Mission Control through the Gateway frame, rather than rebuilding the screens as a custom React interpretation. This allows the owner/designer instruction that the design files are the visual contract and should not be redesigned.

This closes the developer-side trapped-frame defect. It is not owner visual GO until the owner confirms the fixed production UI in the authenticated browser.

Inventory Findings

Files and surfaces inspected:

- src/components/gateway/DesignerGatewayMockFrame.tsx
- src/proxy.ts
- src/app/gateway/page.tsx
- src/app/gateway/agent-hub/page.tsx
- src/app/gateway/agent-hub/paperclip/page.tsx
- src/app/gateway/routes/page.tsx
- src/app/gateway/registry/page.tsx
- src/app/gateway/policies/page.tsx
- src/app/gateway/health/page.tsx
- src/app/gateway/dispatcher/page.tsx
- src/app/gateway/token-governor/page.tsx
- src/app/gateway/bridge-session/page.tsx
- src/app/gateway/node-detail/page.tsx
- src/app/gateway/mobile-tablet/page.tsx
- public/designer-mission-control/design/gateway/*.html
- public/designer-mission-control/design/gateway/shared/tokens.css
- public/designer-mission-control/design/gateway/shared/render.js

Accepted direct mock routes remain:

- '/gateway' -> 'Gateway Overview.html'
- '/gateway/routes' -> 'Gateway Routes.html'
- '/gateway/registry' -> 'Gateway Registry.html'
- '/gateway/policies' -> 'Gateway Policies.html'
- '/gateway/health' -> 'Gateway Health.html'
- '/gateway/dispatcher' -> 'Dispatcher.html'
- '/gateway/token-governor' -> 'Token Governor.html'
- '/gateway/agent-hub' -> 'Agent Hub.html'
- '/gateway/agent-hub/paperclip' -> 'Paperclip.html'
- '/gateway/bridge-session' -> 'Bridge Session Flow.html'
- '/gateway/node-detail' -> 'Gateway Node Detail.html'
- '/gateway/mobile-tablet' -> 'Gateway Mobile Tablet.html'

Root Cause Found

The previous scroll/navigation work removed the outer 'h-screen overflow-hidden' trap, but brows

er proof still showed the Agent Hub frame stuck at the default iframe height of 150px.

Two precise causes were found:

1. Designer asset route framing was blocked:
 - '/designer-mission-control/...' returned 'X-Frame-Options: DENY'
 - its CSP used 'frame-ancestors 'none''
 - effect: the Gateway iframe could not reliably render the same-origin designer asset.
2. The iframe height sync missed already-loaded frames:
 - when the iframe loaded before the React effect listener attached, no height sync ran
 - effect: the iframe stayed at 150px even when the mock document was much taller.

What Was Implemented

- Kept the direct designer mock mounting path.
- Allowed only protected designer assets to be framed by same-origin Gateway pages:
 - designer route: 'X-Frame-Options: SAMEORIGIN'
 - designer route CSP: 'frame-ancestors 'self''
- Kept normal Mission Control/Gateway pages protected from framing:
 - app pages still return 'X-Frame-Options: DENY'
 - app page CSP still uses 'frame-ancestors 'none''
- Added an immediate iframe height sync path when the iframe document is already loaded:
 - checks 'iframe.contentDocument?.readyState === 'complete''
 - calls the same load handler and ResizeObserver setup
 - preserves 'scrolling='yes''
- Added a regression guardrail:
 - exact FULL v3 designer files are mounted by route
 - no return to 'h-screen w-full overflow-hidden'
 - iframe height sync, ResizeObserver, and exits remain present
 - designer assets stay same-origin frameable while app pages stay frame-denied.

Files Changed

- src/components/gateway/DesignerGatewayMockFrame.tsx
- src/proxy.ts
- src/proxy.test.ts
- src/lib/gateway-native-frame-decision.test.ts

No designer HTML, shared tokens, shared classes, '.env', credentials, governance files, or packaged artifacts were changed.

Runtime / Browser Proof

Committed source:

- 25f577e

Standalone deployment:

- Build/deploy script rebuilt committed HEAD 25f577e.
- The deploy script reported:
 - deployed commit: 25f577e
 - pid: 47142
 - port: 3337
- The deploy child exited before route smoke, matching the known standalone child-exit behavior in this local proof environment.
- Controlled proof runtime was started in a local-only screen-backed process:
 - host: 127.0.0.1
 - port: 3337
 - pid: 47252

Browser proof:

- route: '/gateway/agent-hub'
- viewport: 1480x900
- auth method: local proof cookie only, no owner secrets
- iframe source: '/designer-mission-control/design/gateway/Agent%20Hub.html'
- body scroll height: 12539
- iframe height: 12484
- iframe inline height: '12484px'
- iframe ready state: complete
- iframe body height: 12572
- page can scroll: true
- wheel scroll result: 'scrollY=700'
- max scroll: 12519
- exits visible:
 - Mission Control Home: true
 - Gateway Overview: true
 - Agent Hub: true
- designer asset headers:
 - 'X-Frame-Options: SAMEORIGIN'
 - CSP contains 'frame-ancestors 'self''
- Gateway page header:
 - 'X-Frame-Options: DENY'

Result: PASS.

Route Smoke

Smoke base: http://127.0.0.1:3337

Unauthenticated route behavior:

- '/login': 200
- '/gateway': 307 to '/login'
- '/gateway/agent-hub': 307 to '/login'
- '/gateway/agent-hub/paperclip': 307 to '/login'
- '/gateway/dispatcher': 307 to '/login'
- '/gateway/token-governor': 307 to '/login'
- '/gateway/bridge-session': 307 to '/login'
- '/agents': 307 to '/login'
- '/agent-network': 307 to '/login'

Protected action invariant:

- scripts/check-protected-actions-locked.mjs against http://127.0.0.1:3337
- ok: true
- checked: 11
- protected actions remained locked.

Tests Run

- pnpm test src/proxy.test.ts src/lib/gateway-native-frame-decision.test.ts src/lib/mission-control-shell.test.ts
 - 3 files passed
 - 12 tests passed
- git diff --check
 - passed
- pnpm run typecheck
 - passed
- pnpm run build
 - passed
- pnpm test
 - 147 files passed
 - 1296 tests passed
- node scripts/check-protected-file-invariants.mjs
 - ok: true

- staged secret scan
- ok: true
- checked files: 4
- .env diff check
- clean

Safety Confirmation

- No '.env' changes.
- No secrets printed.
- No auth weakening.
- No public local exposure added.
- No fake GO.
- No fake owner visual proof.
- No fake buttons added.
- No designer mock redesign.
- No Gateway FULL v3 token/class changes.
- No parked artifacts staged.
- Designer assets are frameable only by same-origin pages.
- Main Mission Control/Gateway app pages remain protected from external framing.

Remaining Blocker

Blocker: owner_authenticated_gateway_frame_retest_required

Classification: OWNER_GATED

Owner retest needed:

- Log in to Mission Control.
- Open '/gateway'.
- Open '/gateway/agent-hub'.
- Confirm the FULL v3 mock renders rather than a blank/static frame.
- Scroll up/down normally.
- Confirm Mission Control Home, Gateway Overview, and Agent Hub exits are visible.
- Confirm browser back and visible exits are usable.

Closeout Ledger

- Day number and lane: Day 15 - Gateway Native Implementation Decision
- Status: DEVELOPER-SIDE CLOSED
- Blocker classification: OWNER_GATED
- Commit hash: 25f577e
- Push result: pushed to origin/to-knowledge-mc
- Deployed commit: 25f577e
- Runtime proof: local-only standalone proof on 127.0.0.1:3337, pid 47252
- Rollback command: git revert 25f577e
- Next day automatically started: Day 16 - Gateway Designer Fidelity 100% Closure